BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Programming: Subprograms and Parameters

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. How can a loop total an array's values?

2. Why is a thermostat controller an embedded system?

A) It has a specific task
B) It runs any desktop app
C) It always needs Wi-Fi
D) It always has a screen

3. "DNS assigns the same MAC address to all computers visiting a website." Correct this. Explain why.

4. RAM is full. Apps keep moving data between RAM and an SSD. Correct both claims. Explain why. (a) Virtual memory needs no secondary storage. (b) Virtual memory is faster than RAM because SSDs hold more.

Today's objectives

Explain the core ideas of programming: subprograms and parameters using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

Procedure
Runs a named group of instructions, often for a useful action.

Function
Returns a value that can be used in another expression.

Parameter
Allows a subprogram to receive values from its caller.

Modularity
Separating repeated tasks into subprograms reduces duplicated code.

Write one key term and its meaning in your own words:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New learning - Key ideas

Procedure	Runs a named group of instructions, often for a useful action.
Function	Returns a value that can be used in another expression.
Parameter	Allows a subprogram to receive values from its caller.
Modularity	Separating repeated tasks into subprograms reduces duplicated code.

I Do - One worked example

Trace cube(3) for a function returning n*n*n.

1. Caller passes argument 3.
2. Parameter n receives 3.
3. Compute 3 x 3 x 3=27.
4. Return 27 to the caller.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. A procedure printGreeting(name) displays 'Hello '+name for Mia.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete the statement that returns the new two-item array.

```
function pickEnds(values)
array pair[2]
pair[0] = values[0]
pair[1] = values[2]
...
endfunction
array sizes = [2,4,8]
result = pickEnds(sizes)
print(result[1])
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. Explain one difference between a function and a procedure. [2 marks]

Q3. State the role of parameters. [2 marks]

Q4. Give one benefit of modular programming. [2 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. Functions can return values to their callers. ☐ T ☐ F
2. Runs a named group of instructions, often for a useful action. ☐ T ☐ F
3. Returns a value that can be used in another expression. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 Set a total to 0. Loop through the array, adding each value once. ☐

Q2 A - It has a specific task ☐

Q3 Distributed DNS servers help look up the IP address associated with a domain name. ☐

Q4 (a) Virtual memory uses secondary storage when RAM is too full. (b) Moving data between RAM and slower secondary storage takes time. More storage space does not mean faster access. ☐

Code First - Model

```
function pickEnds(values)
array pair[2]
pair[0] = values[0]
pair[1] = values[2]
return pair
endfunction
array sizes = [2,4,8]
result = pickEnds(sizes)
print(result[1])
```

Output: 8

We Do - Check the model

1. Output 'Hello Mia'. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 A function returns a value; a procedure performs statements without necessarily returning a value. ☐

Q3 They pass input values into a subprogram. ☐

Q4 Reduces duplication, simplifies testing or maintenance. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: